

FORECASTING WITH A WORLD MODEL

YOU OBSERVE: TWO HEADS IN TWO FLIPS AT
A CARNIVAL GAME



FORECAST: What is $P(\text{heads})$ on the next coin flip?

DIFFERENT APPROACHES

$P(\text{heads})$

NAÏVE: Assume a fair coin; the setting is just noise.

0.5

FREQUENTIST: Consider only the two flips you saw.

1.0

WORLD MODEL HYPOTHESIS: Carnival coins can be rigged, but two heads could likely be a fair coin. *uses world knowledge and the two flips*

≈ 0.65

What do LLMs Do?

LLMs appear to use an inductive world model to engage in judgemental forecasting.

0.5
NAÏVE

0.65



1.0
FREQUENTIST